
UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-language models (VLMs) have achieved substantial gains on visual question answering benchmarks that span visual reasoning, document understanding, and multimodal dialogue. These gains appear across VLM families that differ in base language model, alignment architecture, and training corpus composition. However, recent evidence indicates that such systems still underperform on conventional image classification benchmarks that probe fine-grained visual knowledge. We evaluate a broad set of recent VLMs on fine-grained classification benchmarks and analyze factors that may explain the discrepancy between fine-grained recognition and aggregate performance on other vision-language benchmarks. Across a sequence of controlled ablation experiments, we find that replacing the LLM improves all benchmark groups at comparable rates, whereas improving the vision encoder yields disproportionate gains on fine-grained classification. We further find that the pretraining stage is a critical determinant of fine-grained performance, particularly when the language-model parameters are unfrozen during pretraining. These results specify concrete design variables for improving fine-grained visual understanding and other vision-centric capabilities in VLMs.

1 INTRODUCTION

Recent progress in vision-language models (VLMs), which couple visual encoders with large language models (LLMs), has produced systems with increasingly strong capacity for interpreting and reasoning over visual inputs (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b). These models support tasks including open-ended visual question answering, document understanding, and multimodal dialogue (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). This progress raises a more specific technical question: to what extent do current VLMs retain fine-grained visual perception capability, and which modeling or training choices improve that capability?

Fine-grained visual recognition—the ability to separate visually similar categories—requires a model to encode subtle discriminative features while remaining invariant to intra-class changes in pose, illumination, and background. Conventional vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have been extensively evaluated on fine-grained classification benchmarks and often perform strongly, but the corresponding behavior of newer LLM-based VLMs has received comparatively limited analysis.

Fine-grained visual knowledge is important because perceptual identification is a prerequisite for many downstream reasoning operations. Deployed VLMs frequently encounter long-tailed visual distributions in which correct behavior depends on distinguishing closely related categories. For example, a model that fails to identify a mushroom species cannot reliably use textual knowledge to infer toxicity or edibility. Consequently, understanding which factors control fine-grained knowledge acquisition can inform architectural and training choices for future VLMs.

We first conduct a systematic evaluation of fine-grained visual knowledge in state-of-the-art VLMs and compare these results with performance on general VLM tasks. Concretely, we evaluate 15 VLMs, including LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al., 2024), on four fine-grained classification benchmarks: ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and

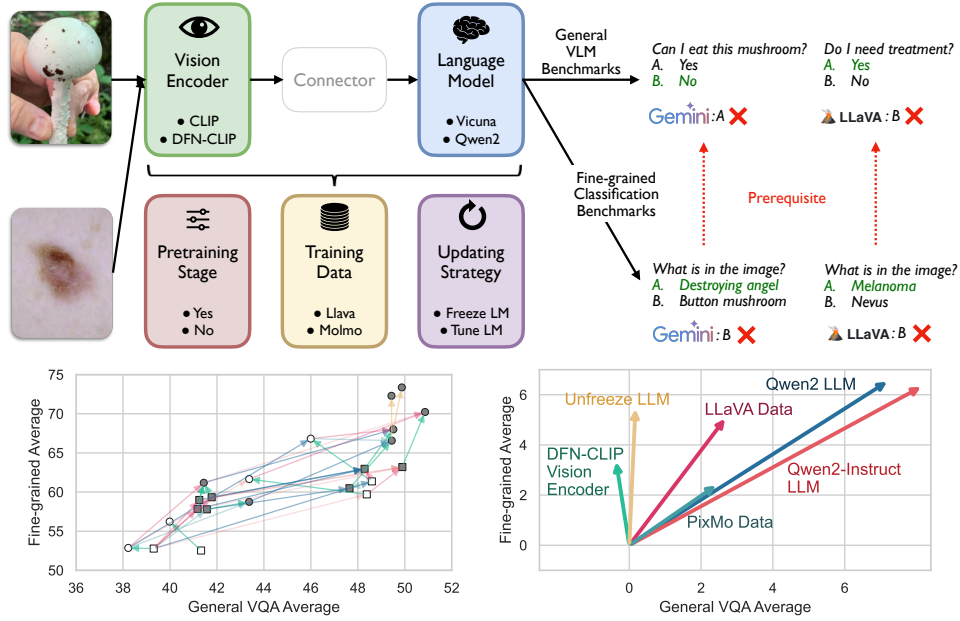


Figure 1: **Overview.** (Top) We evaluate the fine-grained classification capability of vision-language models (VLMs), a vision-centric property that supports higher-level visual understanding and reasoning. (Bottom) Using 22 controlled ablation experiments over model components and training procedures (left), we isolate factors that determine fine-grained classification performance in VLMs (right).

Food (Bossard et al., 2014). We compare these scores with aggregated results from eight general VLM benchmarks, including MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). The results show substantial variation in fine-grained accuracy among models with similar general VLM scores, indicating that fine-grained recognition is a distinct axis of visual capability that is not fully measured by existing VLM benchmarks. We also observe a persistent performance gap between VLMs and their underlying CLIP-based vision encoders, consistent with prior findings (Zhang et al., 2024; Geigle et al., 2024).

To identify mechanisms that affect fine-grained classification in VLMs, we perform a controlled ablation study over model components and training paradigms. We test whether language models and vision encoders contribute differently to classification accuracy, and whether pretraining, data quality, and parameter-freezing choices alter the resulting behavior. Across 22 ablation experiments, we identify several determinants of fine-grained knowledge acquisition in VLMs:

- **Language model (§4.2):** Increasing the capability of the base language model improves both fine-grained classification benchmarks and general VLM benchmarks with similar magnitude.
- **Vision encoder (§4.2):** A stronger vision encoder yields larger gains on fine-grained classification than on general VLM benchmarks, especially for models trained with a two-stage procedure.
- **Pretraining stage (§4.3):** Large-scale image-caption pretraining substantially improves fine-grained benchmark accuracy and is especially effective when the LLM is updated rather than kept frozen.
- **Finetuning stage (§4.3):** Instruction finetuning contributes less to fine-grained classification than base-model selection and pretraining strategy.

Collectively, these findings, including the training-stage analysis in §4.3, indicate that general visual reasoning and fine-grained recognition depend on overlapping but non-identical components of the VLM pipeline. Stronger LLMs improve broad benchmark performance, whereas vision-encoder quality and pretraining dynamics are more directly tied to fine-grained visual knowledge. The resulting design implications support the development of more vision-centric VLMs and transfer naturally to long-tail applications that require fine-grained visual recognition.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) denote multi-modal systems that map visual inputs into the representational space of a language model. Many recent systems instantiate the LLaVA-style architecture (Liu et al., 2023), which consists of three principal modules: a vision encoder, a language model, and an MLP connector that projects visual features into the LLM input space. These systems obtain strong results on multimodal tasks including visual question answering (VQA), document understanding, and general visual reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). Their evaluation has therefore relied on benchmarks such as MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). These benchmarks primarily measure reasoning and language-conditioned answer generation from visual inputs, and they provide limited direct measurement of vision-centric capabilities such as object recognition and fine-grained classification. Our work addresses this measurement gap by systematically evaluating and analyzing VLMs on fine-grained classification, a core but under-specified component of visual intelligence.

Fine-grained visual classification. Fine-grained classification is a standard computer vision problem in which the model must distinguish visually similar subcategories within a broader semantic category, such as bird species (Wah et al., 2011), flower species (Nilsback & Zisserman, 2008), or pet breeds (Parkhi et al., 2012). Benchmarks of this type have been central to training and evaluating vision encoders, including ResNet (He et al., 2016), CLIP (Radford et al., 2021), and DINO (Caron et al., 2021). Although these datasets are important in vision research, they are often omitted from recent VLM evaluation protocols, which instead emphasize visual question answering and multi-modal reasoning. Because deployed systems may operate over long-tail visual distributions and require precise category identification, as in medical diagnosis, food safety, and species identification, fine-grained evaluation is necessary for characterizing VLM usability from a vision-centric perspective rather than only from a language-centric reasoning perspective.

Evaluation and ablation of VLMs for fine-grained classification. Prior work has begun to evaluate VLMs on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024), but these studies typically cover a limited set of models or datasets. We extend this line of work by evaluating a broader set of VLMs across multiple fine-grained classification benchmarks under a unified multiple-choice protocol. We further complement observational evaluation with controlled ablations that isolate factors affecting fine-grained classification performance.

Several recent studies analyze the VLM design space, including model architecture (McKinzie et al., 2024), training strategy (Karamcheti et al., 2024), and multimodal data composition (Gadre et al., 2023; Udandara et al., 2024; Deitke et al., 2024). However, these analyses primarily optimize or report performance on general benchmarks that emphasize language use and reasoning. Here, we examine how base-model selection, pretraining procedure, pretraining data, and instruction-tuning data affect fine-grained classification, thereby specifying design choices that are most relevant for vision-centric VLM behavior.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

This section specifies the motivation for fine-grained classification evaluation, defines the benchmark suite and evaluation protocol, and reports the principal observations from testing 15 existing VLMs.

3.1 MOTIVATION

Fine-grained classification is a core computer vision task in which a model must discriminate among visually similar subcategories within a broader category, such as bird species or car models (Wei et al., 2021). Benchmarks such as ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008) have therefore served as standard testbeds for both training and evaluating vision encoders. With large-scale self-supervised and supervised learning, vision encoders including CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have reached near-human or superhuman performance on challenging fine-grained classification tasks.

These same benchmarks are rarely included in the evaluation protocol for vision-language models (VLMs). Recent VLMs such as LLaVA (Liu et al., 2023) show strong performance in interpreting

visual concepts, but they are typically evaluated on visual question answering (VQA), including academic VQA benchmarks such as MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or document-understanding benchmarks such as DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). Such benchmarks emphasize reasoning, language use, and answer generation, but they contain limited coverage of fine-grained object categories; as a result, they leave core vision-centric recognition behavior largely unmeasured.

Fine-grained classification is operationally important for VLMs because real-world inputs often follow long-tailed distributions and require precise visual discrimination among similar categories. For example, if a blind user asks whether a mushroom is safe to eat, the model must first identify the species before applying textual knowledge about toxicity; confusing a deadly destroying angel with an edible white button mushroom could cause severe harm. Analogously, a medical system that fails to separate visually similar diseases may recommend an incorrect treatment, and an autonomous vehicle that confuses a “stop” sign with a “do not enter” sign may execute an unsafe navigation decision. These examples motivate direct measurement of fine-grained classification as a prerequisite for reliable VLM deployment.

3.2 BENCHMARKS

We measure fine-grained visual knowledge using four established object-recognition benchmarks that cover different recognition domains: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014).

These benchmarks contain between 37 and 1,000 classes, which makes direct multiple-choice evaluation difficult because VLMs are commonly trained and evaluated with a small number of answer options. We therefore follow the procedure of Geigle et al. (2024) and convert each dataset into 5-way multiple-choice questions. For each image, an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) ranks candidate labels by image-label cosine similarity; we select the four highest-scoring incorrect labels as hard negatives and combine them with the ground-truth label to form the answer set. We use OpenCLIP to avoid constructing negatives with the same model family used by VLMs that rely on OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023), all of which are trained on different datasets from OpenCLIP. This conversion places VLMs in their native multiple-choice regime while preserving a large fraction of the original classification difficulty. For example, CLIP ViT-L/14-336px (Radford et al., 2021) obtains 75.2% accuracy on the original 102-class Flowers-102 benchmark and 78.3% accuracy on our 5-way multiple-choice version.

3.3 TESTING SETUP

We evaluate 15 recent VLMs in the 7B–13B parameter range on the converted 5-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abidin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b). For each model, we use its default VLMEvalKit prompt format (Duan et al., 2024) and score a response as correct when it exactly matches the corresponding multiple-choice option. As a baseline, we evaluate CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) with the standard zero-shot protocol, in which images and class-label prompts (“A photo of a [class]”) are encoded by the CLIP image and text encoders. Top-1 classification is obtained by selecting the label with maximum cosine similarity to the image embedding.

To determine whether existing VLM benchmarks measure the same capability, we compare fine-grained classification accuracy with each model’s average score on eight general VQA benchmarks from VLMEvalKit, including MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024).

3.4 RESULTS

Figure 2 plots fine-grained classification performance, averaged across the four benchmarks, against general VQA performance, averaged across eight existing VQA benchmarks, for all evaluated VLMs. Models with similar general VQA scores exhibit large differences in fine-grained classification accuracy. For example, CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu

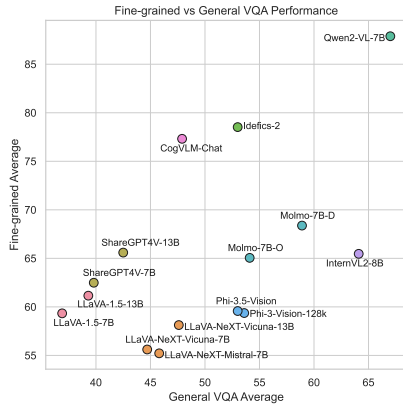


Figure 2: **Fine-grained classification versus general VQA performance across VLMs.** The distribution of recent VLMs indicates that fine-grained classification is a distinct visual capability that is not adequately measured by standard VQA benchmarks.

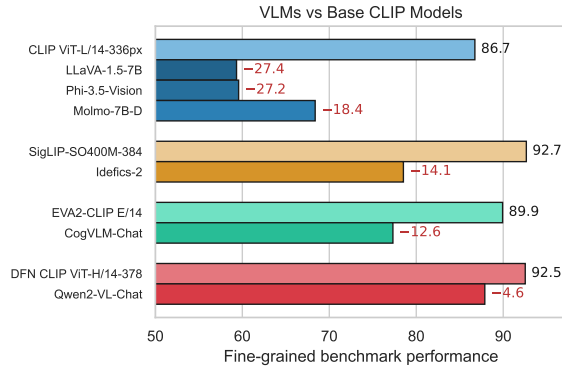


Figure 3: **Comparison between VLMs and their corresponding CLIP vision encoders on fine-grained classification.** Qwen2-VL-Chat nearly matches its DFN-CLIP vision encoder, whereas the other evaluated VLMs remain substantially below their encoder base-lines, indicating remaining headroom for fine-grained classification.

et al., 2024a) both achieve approximately 48% average accuracy on the general VQA benchmarks, but their fine-grained accuracies differ by 19pp (percentage points), with 77.3% for CogVLM-Chat and 58.1% for LLaVA-NeXT-Vicuna-13B. This discrepancy shows that current VLM evaluation benchmarks do not reliably capture fine-grained classification capability, supporting its treatment as a separate dimension of visual intelligence.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

Most current VLMs are constructed on top of a pretrained vision encoder. The multiple-choice formulation therefore permits a direct comparison between each VLM and CLIP-style baselines on the same classification instances. Figure 3 compares selected VLMs with their corresponding CLIP vision encoders and shows a substantial performance gap. For instance, Molmo (Deitke et al., 2024), a recent model with strong VQA performance, reaches 68.4% average accuracy under our multiple-choice evaluation, whereas its CLIP ViT-L/14 (Radford et al., 2021) vision encoder reaches 86.7%. Even Qwen2-VL-Chat (Wang et al., 2024b), the strongest evaluated model, remains 4.6 points below its DFN-CLIP (Fang et al., 2023) vision encoder. This gap motivates a controlled analysis of VLM design choices that may preserve or improve fine-grained visual information during multimodal integration. The next section therefore studies these factors through ablation experiments.

Finding 2: A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

Building on the observational results in Section 3, we ablate model and training factors that may determine fine-grained classification performance, with primary emphasis on base-model selection and training procedure. All ablations use the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023) unless otherwise stated. Each model is trained for one epoch of pretraining, when that stage is included, followed by one epoch of finetuning. We keep the default LLaVA-1.5 hyperparameters, including learning rate and batch size, fixed across experiments to isolate the effects of the ablated variables.

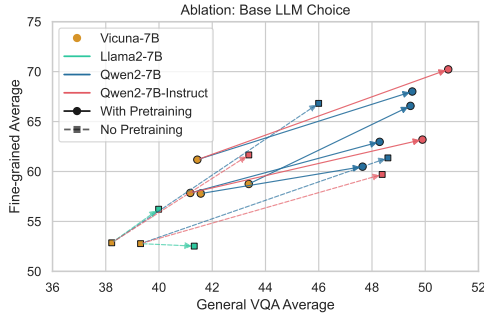


Figure 4: **Base-LLM ablation replacing Vicuna-7B in LLaVA with alternative LLMs.** Averaged across configurations, replacing Vicuna with Qwen2-7B increases fine-grained performance by +7.5pp and general VQA performance by +7.7pp.

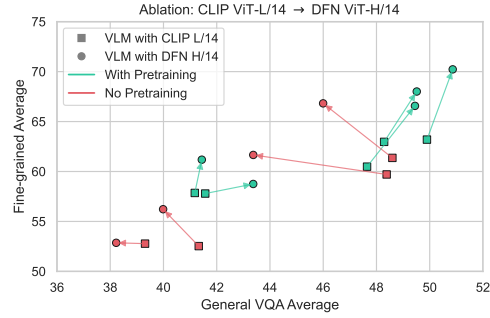


Figure 5: **Vision-encoder ablation replacing CLIP ViT-L/14 in LLaVA with DFN-CLIP ViT-H/14.** With sufficient pretraining, switching to DFN-CLIP improves fine-grained performance by +4.5pp and general performance by +1.2pp.

Fine-grained performance is defined as the average score over the four classification benchmarks in Section 3.2. For general VQA performance, we report the average score on three commonly used multiple-choice benchmarks: MMMU (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b). These benchmarks are selected because they cover multiple domains and are widely used in VLM evaluation.

4.2 BASE MODELS

The analysis in Section 3 shows that recent VLMs often replace the base LLM with newer models while retaining CLIP ViT-L/14 from Radford et al. (2021) as the vision encoder. We therefore isolate how the LLM and vision encoder choices affect fine-grained classification.

4.2.1 LLM CHOICE

Setup. Starting from LLaVA-1.5-7B (Liu et al., 2023), we replace the LLM component while varying vision encoder and pretraining configuration. We compare the original Vicuna-7B-1.5 (Chiang et al., 2023), which is used in LLaVA, with Qwen2-7B (Yang et al., 2024), which is used in Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024). We also evaluate Llama-2-7B (Touvron et al., 2023), the base model underlying Vicuna, and Qwen2-7B-Instruct, the instruction-tuned variant of Qwen2-7B.

Findings. As shown in Figure 4, replacing Vicuna-7B with Qwen2-7B improves performance in all configurations, with mean gains of +7.5pp (percentage points) on fine-grained benchmarks and +7.4pp on general VQA benchmarks. Qwen2-7B-Instruct produces comparable gains of +7.5pp and +8.1pp, respectively. Llama2-7B, the non-instruction-tuned base model for Vicuna, yields smaller improvements, although the downstream consequences of removing instruction tuning are not fully determined by this experiment. All trained models, including the lower-performing Vicuna-based VLMs, consistently emit multiple-choice responses in the expected format. Thus, the observed performance differences are more consistent with differences in option-level visual or semantic knowledge than with differences in output-format compliance.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

4.2.2 VISION ENCODER CHOICE

Setup. Most evaluated VLMs, including the LLaVA family and Molmo, use OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) as the vision encoder. By contrast, Qwen2-VL (Wang et al., 2024b) uses DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which has stronger zero-shot object-

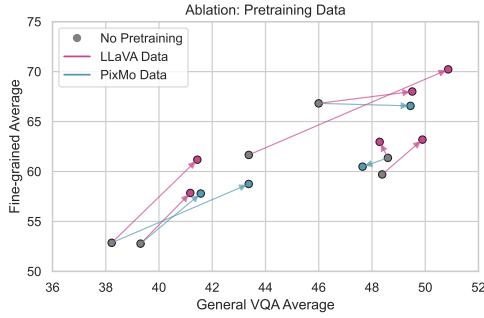


Figure 6: **Pretraining-data ablation.** Adding either LLaVA (CC-3M) or Molmo (PixMo-Cap) pretraining data yields similar gains: LLaVA data increases fine-grained and general scores by +4.0pp and +2.1pp, while PixMo data increases them by +2.4pp and +2.5pp.

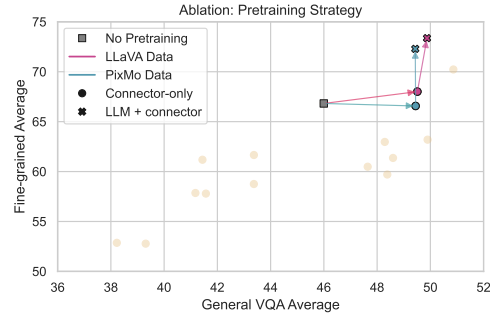


Figure 7: **Training-method ablation comparing connector-only pretraining with LLM+connector tuning.** Unfreezing the LLM during pretraining improves fine-grained performance by +5.5pp without reducing general VQA scores.

recognition performance on benchmarks such as ImageNet. We compare these two encoders across multiple LLM choices and pretraining configurations.

Findings. Figure 5 shows that the effect of replacing CLIP L/14 with DFN-CLIP H/14 depends on whether connector pretraining precedes finetuning. Without pretraining, DFN-CLIP decreases general VQA performance by -2.5pp while increasing fine-grained benchmark scores by +2.8pp. After connector pretraining, general VQA performance is stable or slightly higher (+1.2pp), whereas fine-grained performance increases by +4.5pp. This pattern indicates that higher-capacity vision encoders improve fine-grained classification most reliably when multimodal pretraining has first adapted the connector to the encoder representation.

Takeaway 2: Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pretraining before the finetuning stage.

4.3 TRAINING METHODS

Recent VLMs are commonly optimized in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). The pretraining stage uses large-scale image-caption data and a captioning objective to align the vision encoder with the LLM input space. The instruction-tuning stage then optimizes the VLM on curated multi-round multimodal instruction-completion data to improve response behavior for user-facing tasks. We evaluate how these stages affect fine-grained VLM capability.

4.3.1 PRETRAINING

Setup. We evaluate three questions about the pretraining process:

- Is pretraining required for LLaVA-architecture models when the base components have already been pretrained on billion-scale data?
- How does connector-only pretraining compare with pretraining that also unfreezes the LLM?
- How strongly does pretraining-data quality affect downstream performance?

We test three pretraining-data configurations: LLaVA pretraining data (Liu et al., 2023), consisting of a CC-3M subset with web-scraped captions; Molmo pretraining data (Deitke et al., 2024), namely PixMo-Cap with detailed human-annotated captions; and no pretraining. In most experiments, only the connector is updated during pretraining, following Liu et al. (2023). We additionally evaluate a procedure similar to Deitke et al. (2024): the connector is trained alone for the first 20% of pretraining steps, after which the LLM is unfrozen for the remaining 80%. All pretraining configurations are followed by one epoch of finetuning on LLaVA finetuning data.

Table 1: **Summary of principal ablation settings and comparisons to existing models (bolded).** Improvements to the base LLM, the vision encoder, and joint connector–LLM pretraining on sufficient data increase fine-grained performance, whereas instruction finetuning has a smaller effect. In contrast, general VQA performance is affected more strongly by the LLM choice.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

Findings. Figure 6 reports the pretraining ablation results. Connector-only pretraining usually improves VLM performance, except for the Qwen2 and CLIP ViT-L/14 configuration. The effect is largest for Vicuna-based VLMs, which obtain average gains of +6.1pp on fine-grained classification and +3.1pp on general VQA.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

The difference between web-scraped LLaVA captions and detailed human-annotated PixMo captions is small under connector-only pretraining. Relative to LLaVA data, PixMo data changes fine-grained performance by -1.6pp and general VQA performance by +0.4pp.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance.

One possible explanation is that connector-only pretraining restricts the model’s ability to use richer caption supervision because the LLM parameters remain fixed. To test this explanation, we pre-train both the LLM and connector after a 20% connector-only warmup phase. Figure 7 shows that this procedure increases fine-grained classification performance by +5.5pp without reducing general VQA performance. The same improvement appears with both LLaVA and PixMo pretraining data, indicating that the lower-quality LLaVA captions do not measurably degrade the LLM’s language capabilities in this setting.

Takeaway 5: Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

4.3.2 FINETUNING

Setup. For this ablation, we start from the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b), which is the base model used to train Qwen2-VL-7B-Chat, and finetune it on LLaVA-Instruct data (Liu et al., 2023) using the same protocol as above. We compare the resulting model with Qwen2-VL-7B-Chat, which uses 2M in-house examples for instruction finetuning (Wang et al., 2024b). This comparison isolates the contribution of the instruction-finetuning dataset while holding the pretrained architecture fixed.

Findings. Finetuning only on LLaVA finetuning data slightly reduces both fine-grained and general benchmark scores. Fine-grained accuracy decreases by 2.4pp, from 87.9% to 85.5%, and general benchmark performance decreases by 1.3pp, from 62.4% to 61.1%. As shown in Figure 8, this is the smallest fine-grained contribution among the studied factors, suggesting that instruction finetuning is less central to fine-grained knowledge acquisition than base-model selection and pretraining procedure.

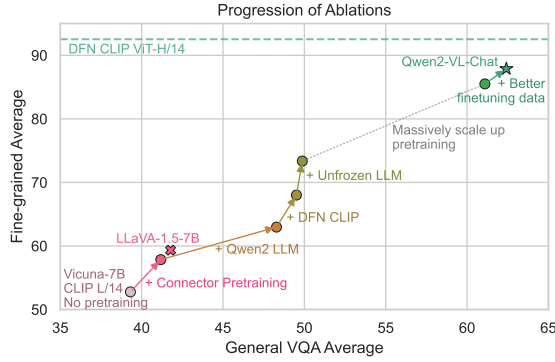


Figure 8: **Progression of combined ablation effects.** Starting from Vicuna-7B with CLIP ViT-L/14, successive changes to the base LLM, vision encoder, and training configuration increase fine-grained classification accuracy from 52.8% to 73.4%, closing more than half of the gap to Qwen2-VL-Chat’s 87.9%. The remaining gap is primarily attributed to Qwen2-VL’s 1.4T-token pretraining corpus.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 summarize how the ablated components reduce the performance gap between weaker and stronger VLMs on fine-grained and general VQA tasks. Starting from the baseline LLaVA architecture (Liu et al., 2023), with Vicuna-7B (Chiang et al., 2023) and CLIP ViT-L/14 (Radford et al., 2021), the sequence of component changes increases fine-grained performance from 52.8% to 73.4% and general VQA performance from 39.3% to 49.9%. The ablation trajectory indicates that replacing the vision encoder with DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining preferentially improve fine-grained capability, whereas replacing the LLM with Qwen2-7B (Yang et al., 2024) explains most of the general VQA gain.

Even after these changes, Table 1 leaves an approximately 12-point fine-grained classification gap that is not explained by the ablations. We consider two plausible contributors to this residual gap: architectural differences and pretraining data scale.

Although architectural changes may contribute, the observational results in Section 3.4 and the training ablations in Section 4.3 suggest that pretraining data scale is the dominant remaining factor. Our LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) pretraining experiments use fewer than 1M image-caption pairs, corresponding to approximately 200M and 400M tokens, respectively. In contrast, Wang et al. (2024b) report pretraining Qwen2-VL on 1.4T tokens, which is orders of magnitude larger than our experimental pretraining corpora. This scale difference likely accounts for much of the unexplained performance gap and indicates that extensive pretraining remains an important requirement for VLMs with strong fine-grained classification capability.

5 LIMITATIONS & CONCLUSION

This study has practical limitations that delimit the scope of the conclusions and motivate further work. Because of computational constraints, our training comparisons use fewer than 1M examples rather than the billion-scale or trillion-token pretraining regimes used by some recent VLMs; consequently, the interaction between our identified factors and substantially larger pretraining corpora remains unresolved. In addition, newer work proposes alternative pretraining strategies, and those procedures may interact differently with fine-grained visual understanding than the LLaVA-style training pipeline studied here.

We systematically evaluate state-of-the-art vision-language models (VLMs) on fine-grained classification benchmarks and identify fine-grained visual classification as an important but under-measured dimension of VLM capability. Through controlled analysis of base models, vision encoders, pre-

486 training decisions, and finetuning data, we characterize mechanisms that improve fine-grained clas-
487 sification and strengthen vision-centric behavior. These findings provide actionable guidance for
488 building VLMs intended for real-world settings in which reliable operation depends on precise vi-
489 sual recognition.
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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: Detailed model-ablation results across aggregate evaluation metrics.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMU	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: Detailed model-ablation results on general VQA benchmarks.

A FULL RESULTS

We present the complete ablation results for aggregated performance, general VQA benchmarks, and fine-grained classification in Table 2, Table 3, and Table 4, respectively.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: **Detailed model-ablation results on fine-grained classification benchmarks.**

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

As described in Section 3.2, we evaluate VLMs and their base vision encoders on four conventional classification benchmarks:

ImageNet-1K (Deng et al., 2009): A foundational image-classification dataset derived from the WordNet hierarchy and covering a broad set of supercategories, each of which contains multiple fine-grained subcategories. The ImageNet test set contains 50,000 images distributed over 1,000 classes.

The ImageNet/WordNet hierarchy contains many classes that are visually too distinct from neighboring classes to support hard multiple-choice construction; for example, only three classes fall under the “person” category. Because of this limitation, and because the full dataset is large, we use human-annotated incorrect model predictions from Recht et al. (2019) to identify images for which hard negative answers can be constructed. Starting from 19,056 annotated examples, we filter the pool to 5,457 images covering 928 of the original 1,000 classes. Since this curation produces an imbalanced class distribution, we report mean-per-class accuracy for ImageNet.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): A 102-class flower-species dataset characterized by high intra-class variation and high inter-class visual similarity. The Flowers test set contains 6,149 examples.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): A 37-class pet-category dataset with substantial variation in pose, illumination, and occlusion. The Pets test set contains 3,669 examples.

Food-101 (Bossard et al., 2014): A 101-class food-recognition dataset with substantial within-class visual diversity. The Food test set contains 25,250 examples.

Question generation. For all fine-grained classification test sets except ImageNet, which is handled as described above, we generate hard negatives by applying zero-shot classification to the test examples with an OpenCLIP ViT-L/14 model trained on LAION (Schuhmann et al., 2022), following Geigle et al. (2024). For each instance, we select the top four incorrect predictions and shuffle them with the correct label to produce the answer choices.

Prompt formatting. In the multiple-choice VQA formulation, the question is always “What type of object is in this photo?” The answer options are formatted as “A. Option 1\n B. Option 2\n C. Option 3\n D. Option 4\n E. Option 5”. Because VLMs are trained with different multiple-choice prompts and output conventions, we use the corresponding VLMEvalKit format for each model (Duan et al., 2024); for example, one instruction is “Answer with the option’s letter from the given choices directly.”

For CLIP models, we use the standard zero-shot classification protocol, so no question prompt is required. Class names are instead passed to the text encoder with the template “a photo of a {classname}”.

B.2 GENERAL VQA

For the initial observational evaluation in Section 3, we use the scores reported by the OpenVLM Leaderboard (Duan et al., 2024). The default leaderboard score is the average over eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). This reported average is used only in Figure 2.

For all subsequent evaluations, including existing models and the models trained in Section 4, we restrict the comparison to multiple-choice VLM benchmarks so that the evaluation format more closely matches our fine-grained classification protocol. Specifically, we evaluate VLMs on three benchmarks:

MMBench (Liu et al., 2024b): A 2,948-example dataset that probes a broad set of VLM capabilities, divided between perception and reasoning. Most examples emphasize coarse-grained perception or visual reasoning, although some examples are labeled as “fine-grained perception”; these categories, such as action recognition, attribute recognition, and OCR, differ from the fine-grained classification datasets studied here.

MMMU (Val) (Yue et al., 2024): A large-scale multimodal dataset spanning a wide set of domains and applications. Although the test set contains 10,500 examples, most prior work reports performance on the validation split, which contains 900 questions. MMMU evaluates general knowledge grounded in visual perception across art, science, engineering, medicine, business, and the humanities.

MMStar (Chen et al., 2024b): A 1,500-question dataset selected by human annotators. The examples are chosen to cover diverse domains and low-level model capabilities.

Evaluation procedure. For the general VQA benchmarks, we use the VLMEvalKit evaluation code (Duan et al., 2024) without modification. Existing models use their predefined evaluation procedures, and our trained models use the LLaVA evaluation code because all of them are finetuned on LLaVA-Instruct data.

C TRAINING

We build on the LLaVA-1.5 training codebase (Liu et al., 2023). Most experiments retain the LLaVA architecture and training procedure so that the ablations isolate the effect of the specified model or data variable.

Pretraining. The pretraining stage consists of one epoch on either LLaVA pretraining data, constructed from CC-3M images and captions, or Molmo pretraining data (PixMo-Cap). For PixMo-Cap, we use only the “caption” field and exclude the raw audio transcript. During pretraining, the batch size is 256. In most pretraining runs, we update only the randomly initialized MLP connector with a learning rate of $1e-3$. When ablating pretraining strategy, we first train only the connector for 20% of the steps as a warmup phase, then unfreeze and train the LLM with learning rate $2e-5$ for the remaining 80% of steps.

Finetuning. The finetuning stage always consists of one epoch on the LLaVA-1.5 finetuning data, which combines LLaVA-Instruct with VQA training data from multiple datasets to support bench-

mark evaluation. During finetuning, the LLM learning rate is $2e-5$ and the batch size is 128. The vision encoder remains frozen in both pretraining and finetuning.

For the ablation using the Qwen2-VL architecture, we initialize from Qwen2-VL-7B Base and apply the same training protocol, with only the LLM unfrozen.

Compute. All models are trained on 4 nodes with 4 A100 (40GB) GPUs per node. Training typically completes within 20 hours, with runtime variation determined primarily by CLIP model size.

LLaVA vs. PixMo data. The main text distinguishes LLaVA and PixMo pretraining data by caption quality. LLaVA pretraining data is derived from CC-3M, which contains web-scraped captions that are often low quality or weakly relevant. PixMo-Cap is collected by asking human annotators to record a 60–90 second audio description of an image and then converting that description into an image caption (Deitke et al., 2024). A surface-level dataset analysis shows that LLaVA captions contain 9.8 words on average, whereas PixMo captions contain 169 words on average.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”